Name

Mathematics Advanced Year 11

Logarithmic Functions

Book 1

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Syllabus Content

MAV-11-08 analyses graphs of exponential and logarithmic functions

Logarithmic functions

- Define the logarithm of a number y, where $y > 0$, to any positive base a as the index to which a is raised to give y
- Use the notation $\log_a y$ for the logarithm of y to the base a
- Define the natural logarithm $\ln a = \log_e a$
- Use digital tools to determine rational and irrational values of exponential and logarithmic expressions
- Explain that $y = a^x$ is equivalent to $x = \log_a y$ for $a > 0$ and $a \neq 1$, and use the equivalence to solve equations of the form $a^x = b$, for $a > 0$
- Recognise and use the logarithmic properties: $\log_a a^x = x$ for all real x, $a^{\log_a x} = x$ where $x > 0$
- Derive the laws of logarithms from the laws of indices $\log_a m + \log_a n = \log_a(mn)$, $\log_a m - \log_a n = \log_a\left(\frac{m}{n}\right)$ and $\log_a(m^n) = n\log_a m$
- Justify the logarithmic results $\log_a a = 1$, $\log_a 1 = 0$ and $\log_a\left(\frac{1}{x}\right) = -\log_a x$
- Apply the logarithmic laws and results to simplify expressions, solve equations and prove results, using digital tools where necessary
- Prove the change of base rule $\log_a x = \frac{\log_b x}{\log_b a}$ and use it to solve problems
- Graph the logarithmic function $y = \log_a x$ for $a > 0$ and $a \neq 1$
- Use graphing applications to identify the graphs of $y = a^x$ and $y = \log_a x$ as reflections of each other in the line $y = x$, in cases where $a > 1$ and $0 < a < 1$

MAV-12-02 models, analyses and solves practical problems involving functions and their transformations

Modelling with functions

- Recognise that logarithmic scales are useful to represent a variable that ranges over a large set of values and explain when a logarithmic scale is suitable
- Use logarithmic scales to model and solve problems involving decibels (dB) in acoustics, seismic scales for earthquakes, magnitudes of stars and the pH scale in chemistry

Introduction to Logarithms

Review

☐ Recall powers of numbers.

x	0	1	2	3	4	5
2^x						
3^x						243
4^x					256	
5^x		5				
10^x			100			

☐ Solve exponential equations.

a $2^x = 16$

b $3^x = 81$

c $7^x = 343$

d $5^x = \sqrt{x}$

e $9^x = 3$

f $4^x = \frac{1}{16}$

Investigation Logarithms

- **Roots** find the base when you know the power.

$$x^3 = 8$$
$$\sqrt[3]{\quad} \qquad \sqrt[3]{\quad}$$
$$x = \sqrt[3]{8}$$

- **Logarithms** find the power when you know the base.

$$2^x = 8$$
$$\log_2 \qquad \log_2$$
$$x = \log_2 8$$

Solve these equations using your knowledge of exponentiation and write the logarithm.

Exponential	$3^x = 9$	$3^x = 27$	$3^x = 81$
Logarithm	$x = \log_3 9 =$	$x = \log_3 27 =$	$x = \log_3 81 =$

Exponential	$2^x = 8$	$2^x = 16$	$2^x = 64$
Logarithm	$x = \log_2 8 =$	$x =$	

Exponential	$2^x = 256$	$4^x = 256$	$16^x = 256$
Logarithm			

Exponential	$5^x = 25$	$5^x = \frac{1}{25}$	$6^x = \frac{1}{36}$
Logarithm			

Logarithms are the inverse of exponentiation, they tell you the ……, given a base number.

The log form of $8 = 2^3$ is

The log form of $y = a^x$ is

$\log_3 27$ is read as "log base three twenty-seven." It means "the power of …… that gives 27"

ⓘ Logarithms are the Inverse of Exponentiation

For $a > 0, y > 0$:

$$a^x = y$$
$$x = \log_a y$$

🔍 **Convert** from index form to log form.

Write the following in logarithm form.

$6^2 = 36$ $2 = \log_6 36$	$10^{-2} = \frac{1}{100}$ $-2 = \log_{10} \frac{1}{100}$	$9^{\frac{1}{2}} = 3$ $\frac{1}{2} = \log_9 3$
a $7^2 = 49$	**b** $5^0 = 1$	**c** $10^{-2} = 0.01$
d $16^{\frac{1}{2}} = 4$	**e** $10^4 = 10\,000$	**f** $\left(\frac{3}{4}\right)^2 = \frac{9}{16}$

🔍 **Convert** from log form to index form.

Write the following in index form.

$\log_3 9 = 2$ $9 = 3^2$	$\log_8 2 = \frac{1}{3}$ $2 = 8^{\frac{1}{3}}$	$\log_6 \frac{1}{36} = -2$ $\frac{1}{36} = 6^{-2}$
a $\log_7 49 = 2$	**b** $\log_{49} 7 = \frac{1}{2}$	**c** $\log_5 \frac{1}{25} = -2$
d $\log_8 16 = \frac{4}{3}$	**e** $1 = \log_7 7$	**f** $\log_3 5 = 1.465$

1 Write in index form.

a $\log_2 16 = 4$ — $16 = 2^4$

b $\log_{10} 100 = 2$ — $100 = 10^2$

c $\log_3 27 = 3$ — $27 = 3^3$

d $\log_2 \left(\frac{1}{4}\right) = -2$ — $\frac{1}{4} = 2^{-2}$

e $\log_{10} 0.1 = -1$ — $0.1 = 10^{-1}$

f $\log_3 \left(\frac{1}{9}\right) = -2$ — $\frac{1}{9} = 3^{-2}$

2 Write in log form.

a $2^3 = 8$ — $3 = \log_2 8$

b $3^4 = 81$ — $4 = \log_3 81$

c $2^5 = 32$ — $5 = \log_2 32$

g $4^2 = 16$ — $2 = \log_4 16$

h $10^{-1} = \frac{1}{10}$ — $-1 = \log_{10}\left(\frac{1}{10}\right)$

i $5^{-3} = \frac{1}{125}$ — $-3 = \log_5\left(\frac{1}{125}\right)$

3 Write in log form.

a $3^x = y$ — $x = \log_3 y$

b $5^x = z$ — $x = \log_5 z$

c $x^2 = y$ — $2 = \log_x y$

d $2^b = a$ — $b = log_2 a$

e $b^3 = d$ — $3 = \log_b d$

f $y = 8^x$ — $\log_8 y = x$

g $y = 6^x$ — $\log_6 y = x$

h $y = e^x$ — $\log_e y = x$

i $y = a^x$ — $\log_a y = x$

4 Write in index form.

a $\log_3 5 = x$ — $5 = 3^x$

b $\log_4 7 = x$ — $7 = 4^x$

c $\log_3 a = b$ — $a = 3^b$

d $\log_x y = 9$ — $y = x^9$

e $\log_a b = y$ — $b = a^y$

f $y = \log_2 6$ — $2^y = 6$

g $y = \log_3 x$ — $3^y = x$

h $y = \log_{10} 9$ — $10^y = 9$

i $y = \log_e 4$ — $e^y = 4$

5 Rewrite each equation in index form, then use your knowledge of powers to solve for x.

a $x = \log_{10} 1000$

$10^x = 1000, x = 3$

b $x = \log_{10} 10$

$10^x = 10, x = 1$

c $x = \log_{10} 1$

$10^x = 1, x = 0$

d $x = \log_{10}\left(\frac{1}{100}\right)$

$10^x = \frac{1}{100}, x = -2$

e $x = \log_3 9$

$3^x = 9, x = 2$

f $x = \log_5 125$

$5^x = 125, x = 3$

g $x = \log_2 64$

$2^x = 64, x = 6$

h $x = \log_4 64$

$4^x = 64, x = 3$

i $x = \log_8 64$

$8^x = 64, x = 2$

j $x = \log_7\left(\frac{1}{7}\right)$

$7^x = \frac{1}{7}, x = -1$

k $x = \log_{12}\left(\frac{1}{12}\right)$

$12^x = \frac{1}{12}, x = -1$

l $x = \log_{11}\left(\frac{1}{121}\right)$

$11^x = \frac{1}{121}, x = -2$

m $x = \log_6\left(\frac{1}{36}\right)$

$6^x = \frac{1}{36}, x = -2$

n $x = \log_4\left(\frac{1}{64}\right)$

$4^x = \frac{1}{64}, x = -3$

o $x = \log_8\left(\frac{1}{64}\right)$

$8^x = \frac{1}{64}, x = -2$

6 Solve for x by first rewriting index form, or otherwise.

a $\log_2 64 = x$

6

b $\log_{10} 1000 = x$

3

c $\log_2 x = 4$

16

d $\log_3 x = 4$

81

e $\log_{10} x = 3$

1000

f $\log_3 x = -2$

$\frac{1}{9}$

g $\log_4 x = -1$

$\frac{1}{4}$

h $\log_7 x = -3$

$\frac{1}{343}$

i $\log_x 27 = 3$

3

j $\log_x 32 = 5$

2

k $\log_x 64 = 3$

4

l $\log_x 81 = 4$

3

7 Solve each equation for x

a $x = \log_7\sqrt{7}$

$7^x = \sqrt{7}, x = \frac{1}{2}$

b $x = \log_{11}\sqrt{11}$

$11^x = \sqrt{11}, x = \frac{1}{2}$

c $\log_9 x = \frac{1}{2}$

$x = 9^{\frac{1}{2}} = 3$

d $\log_{144} x = \left(\frac{1}{2}\right)$

$x = 144^{\frac{1}{2}} = 12$

e $\log_x 3 = \left(\frac{1}{2}\right)$

$x^{\frac{1}{2}} = 3, x = 9$

f $\log_x 13 = \frac{1}{2}$

$x^{\frac{1}{2}} = 13, x = 169$

g $x = \log_6\sqrt[3]{6}$

$6^x = \sqrt[3]{6}, x = \frac{1}{3}$

h $x = \log_9 3$

$9^x = 3, x = \frac{1}{2}$

i $\log_{64} x = \frac{1}{3}$

$x = 64^{\frac{1}{3}} = 4$

j $\log_{16} x = \frac{1}{4}$

$x = 16^{\frac{1}{4}} = 2$

k $\log_x 2 = \frac{1}{3}$

$x^{\frac{1}{3}} = 2, x = 8$

l $\log_x 2 = \frac{1}{6}$

$x^{\frac{1}{6}} = 2, x = 64$

m $x = \log_8 2$

$8^x = 2, x = \frac{1}{3}$

n $x = \log_{125} 5$

$125^x = 5, x = \frac{1}{3}$

o $\log_7 x = \frac{1}{2}$

$x = 7^{\frac{1}{2}} = \sqrt{7}$

p $\log_7 x = -\frac{1}{2}$

$x = 7^{-\frac{1}{2}} = \frac{1}{\sqrt{7}}$

q $\log_x\left(\frac{1}{7}\right) = -\frac{1}{2}$

$x^{-\frac{1}{2}} = \frac{1}{7}, x = 49$

r $\log_x\left(\frac{1}{20}\right) = -\frac{1}{2}$

$x^{-\frac{1}{2}} = \frac{1}{20}, x = 400$

j $x = \log_4\left(\frac{1}{2}\right)$

$4^x = \frac{1}{2}, x = -\frac{1}{2}$

k $x = \log_{27}\left(\frac{1}{3}\right)$

$27^x = \frac{1}{3}, x = -\frac{1}{3}$

l $\log_{121} x = -\frac{1}{2}$

$x = 121^{-\frac{1}{2}} = \frac{1}{11}$

m $\log_{81} x = -\left(\frac{1}{4}\right)$

$x = 81^{-\frac{1}{4}} = \frac{1}{3}$

n $\log_x\left(\frac{1}{2}\right) = -\frac{1}{4}$

$x^{-\frac{1}{4}} = \frac{1}{2}, x = 16$

o $\log_x 2 = -\frac{1}{4}$

$x^{-\frac{1}{4}} = 2, x = \frac{1}{16}$

8 A single bacterium cell divides into two every minute.

a Complete this cell population table.

Time (min)	0	1	2	3	4
Population	1	2	4	8	16

b Write an equation for the population, P, after t minutes.

$P = 2^t$

c Use your rule to find the population after 8 minutes.

$P = 2^8 = 256$

d Use trial and error to find the time (correct to the nearest minute) for the population to rise to 10000.

13 minutes

e Write the exact answer to part **d** as a logarithm.

$2^t = 1000$

$t = \log_2 1000$

9 **HSC Sample Question Band 4**

If $y = \log_a(7x - b) + 3$, then x is equal to:

(A) $\frac{1}{7}a^{y-3} + b$

(B) $\frac{y-3}{\log_a(7-b)}$

(C) $\frac{1}{7}\left(a^{y-3} + b\right)$

(D) $a^{y-3} - \frac{b}{7}$

$y - 3 = \log_a(7x - b)$

$a^{y-3} = 7x - b$

$7x = a^{y-3} + b$

$x = \frac{1}{7}(a^{y-3} + b)$

C

The Logarithm Laws

Review

☐ Simplify expressions using index laws.

a	$(\sqrt{3})^2$	**b**	$\sqrt{3^2}$	**c**	$3^x \times 3^y$	**d**	$\frac{3^x}{3^y}$
e	$(3^x)^y$	**f**	3^0	**g**	3^{-1}	**h**	3^{-x}

The Inverse Relationship

Logarithms and exponentiating are inverse operations.

Taking the logarithm will "undo" exponentiation.

$\log_a(a^x) = x$ for all real x

$a^{\log_a x} = x$ for all real $x > 0$

Identify inverse relationship between logarithms and exponentiation.

Simplify

$\log_7 7^{12} = 12$	$\log_{11} 11^{15} = 15$	$3^{\log_3 5} = 5$

a	$\log_{10} 10^3$	**b**	$\log_{12} 12^{1.3}$	**c**	$\log_a a^{14}$
d	$6^{\log_6(-2)}$	**e**	$e^{\log_e 97}$	**f**	$3^{\log_5 3}$

ⓘ The Log Laws

The logarithm laws work like the index laws

Inverse Relationship	$\log_a a^x = a^{\log_a x} = x$
Log Law of Multiplication	$\log_a(xy) = \log_a x + \log_a y$
Log Law of Division	$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$
Log Law of Indices	$\log_a(x^n) = n\log_a x$

🔍 **Simplify** expressions using the log laws.

a	$\log_6 2 + \log_6 4$	**b**	$\log_2 3 + \log_2 7$	**c**	$\log_3 2 + \log_5 2$
d	$\log_6 2 - \log_6 4$	**e**	$\log_2 3 - \log_2 7$	**f**	$\log_3 2 - \log_5 2$
g	$\log_6(2^5)$	**h**	$\log_2(7^{-3})$	**i**	$\log_3(3^5)$
j	$\log_9 18 - \log_9 3$	**k**	$\log_7 2 + \log_7 3$	**l**	$\log_a 22 - \log_a 11$
m	$\log_a(2^3)$	**n**	$\log_3 15 + \log_3 3$	**o**	$\log_4(5^2)$

ⓘ Evaluating Logarithms using Log Laws

$\log_a(1) = 0$	because $1 = a^0$
$\log_a(a) = 1$	because $a = a^1$
$\log_a(\sqrt{a}) = \frac{1}{2}$	because $\sqrt{a} = a^{\frac{1}{2}}$
$\log_a(\sqrt{x}) = \frac{1}{2}\log_a(x)$	because $\log_a(\sqrt{x}) = \log_a(x^{\frac{1}{2}}) = \frac{1}{2}\log_a x$
$\log_a \frac{1}{a} = -1$	because $\frac{1}{a} = a^{-1}$
$\log_a\left(\frac{1}{x}\right) = -\log_a x$	because $\log_a\left(\frac{1}{x}\right) = \log_a(x^{-1}) = -\log_a x$

🔍 **Evaluate** expressions using the log laws.

a	$\log_3 1$	**b**	$\log_7 7$	**c**	$\log_3 \sqrt{3}$
d	$\log_4 \frac{1}{4}$	**e**	$\log_5 \sqrt{7}$	**f**	$\log_6 \sqrt{6}$
g	$\log_{12} \frac{1}{12}$	**h**	$\log_{100} 1$	**i**	$\log_5 \frac{1}{6}$

Example Evaluate expressions using log laws.

$\log_2 6 - \log_2 3$

$= \log_2 2$

$= 1$

$\log_6 \frac{1}{36}$

$= \log_6 36^{-1}$

$= \log_6 6^{-2}$

$= -2 \log_6 6$

$= -2$

Practice

a $\log_6 2 + \log_6 3$

1

b $\log_3 15 - \log_3 5$

1

c $\log_2 16$

4

d $\log_4 \frac{1}{64}$

−3

e $\log_5 25 - \log_5 \sqrt{5}$

1.5

f $\frac{1}{2}\log_2 4 + 2\log_2 8$

7

1 State the missing numbers.

a $\log_2 \square = 1$ — 2

b $\log_3 \square = 0$ — 1

c $\log_a 2 + \log_a \square = \log_a 8$ — 4

d $\log_a 36 - \log_a \square = \log_a 3$ — 12

e $\log_a 3^4 = \square \times \log_a 3$ — 4

f $\square \log_a 3 = \log_a \frac{1}{3}$ — −1

2 Simplify using an appropriate log law.

a $\log_a 7 + \log_a 4$ — $\log_a 28$

b $4 \log_a 2$ — $\log_a 16$

c $2 \log_a 3$ — $\log_a 9$

d $\log_b 7 - \log_b 5$ — $\log_b \left(\frac{7}{5}\right)$

e $\log_a 5 + \log_a 3$ — $\log_a 15$

f $\log_b 15 + \log_b 1$ — $\log_b 15$

g $\log_a 10 - \log_a 5$ — $\log_a 2$

h $3 \log_a 3$ — $\log_a 27$

i $\log_b 28 - \log_b 14$ — $\log_b 2$

j $2 \log_a 5$ — $\log_a 25$

k $\log_b 1 + \log_b 17$ — $\log_b 17$

l $5 \log_a 2$ — $\log_a 32$

m $\log_a 3 + \log_a 2$ — $\log_a 6$

n $\log_a 100 - \log_a 10$ — $\log_a 10$

o $\log_b 6 + \log_b 3$ — $\log_b 18$

p $3 \log_a 10$ — log_a 1000

q $\log_a 36 - \log_a 12$ — $\log_a 3$

r $\log_b 3 - \log_b 2$ — $\log_b \left(\frac{3}{2}\right)$

3 Evaluate:

a $\log_3 1$ — 0

b $\log_7 1$ — 0

c $\log_x 1$ — 0

d $\log_4 4$ — 1

e $\log_{18} 18$ — 1

f $\log_a a$ — 1

g $5 \log_2 1$ — 0

h $3 \log_4 4$ — 3

i $\frac{1}{3} \log_7 7$ — $\frac{1}{3}$

j $\frac{2}{3} \log_{10} 10$ — $\frac{2}{3}$

k $\log_{15} 225$ — 2

l $\log_3 243$ — 5

4 Evaluate the following expressions. The answers are whole numbers.

a	$\log_6 9 + \log_6 4$	**b**	$\log_5 75 - \log_5 3$	**c**	$\log_{15} 3 + \log_{15} 5$
	2		2		1
d	$\log_{12} 72 + \log_{12} 2$	**e**	$\log_{10} 50 + \log_{10} 20$	**f**	$\log_3 15 - \log_3 5$
	2		3		1
g	$\log_2 24 - \log_2 3$	**h**	$\log_3 810 - \log_3 10$	**i**	$\log_2 12 + \log_2 6 - \log_2 9$
	3		4		3
j	$\log_{30} 2 + \log_{30} 3 + \log_{30} 5$	**k**	$\log_{12} 9 + \log_{12} 8 + \log_{12} 2$	**l**	$\log_2 12 + \log_2 6 - \log_2 72$
	1		2		0
m	$\log_5 2 - \log_5 50$	**n**	$\log_2 6 - \log_2 48$	**o**	$\log_3 8 - \log_3 6 - \log_3 12$
	-2		-3		-2
p	$\log_5 12 - \log_5 20 - \log_5 15$	**q**	$\log_6 \left(\frac{1}{3}\right) - \log_6 12$	**r**	$\log_7 \left(\frac{1}{9}\right) + \log_7 63$
	-2		-2		1

5 Write each expression in terms of $\log_a 2$

a	$\log_a 8$	**b**	$\log_a 16$	**c**	$\log_a 64$	**d**	$\log_a \left(\frac{1}{2}\right)$
	$3\log_a 2$		$4\log_a 2$		$6\log_a 2$		$-\log_a 2$
e	$\log_a \left(\frac{1}{8}\right)$	**f**	$\log_a \left(\frac{1}{32}\right)$	**g**	$\log_a \sqrt{2}$	**h**	$\log_a \left(\frac{1}{\sqrt{2}}\right)$
	$-3\log_a 2$		$-5\log_a 2$		$\frac{1}{2}\log_a 2$		$-\frac{1}{2}\log_a 2$

6 Given that $log_2 3 \approx 1.58$ and $log_2 5 \approx 2.32$, use the log laws to find approximations for

a	$\log_2 15$	**b**	$\log_2 9$	**c**	$\log_2 10$	**d**	$\log_2 50$
	3.90		3.16		3.32		5.64
e	$\log_2 \left(\frac{3}{2}\right)$	**f**	$\log_2 \left(\frac{3}{5}\right)$	**g**	$\log_2 \left(\frac{2}{3}\right)$	**h**	$\log_2 75$
	0.58		-0.74		-0.58		6.22

7 Simplify:

a $\log_{10} 10^3$ — 3

b $\log_7 7^5$ — 5

c $\log_{12} 12^{1.3}$ — 1.3

d $\log_8 8^n$ — n

8 Simplify:

a $10^{\log_{10} 100}$ — 100

b $3^{\log_3 7}$ — 7

c $4^{\log_4 36}$ — 3.6

d $6^{\log_6 y}$ — y

9 Evaluate:

a $\log_2\left(\frac{1}{4}\right)$ — -2

b $\log_3\left(\frac{1}{27}\right)$ — -3

c $\log_4\left(\frac{1}{64}\right)$ — -3

d $\log_5\left(\frac{1}{5}\right)$ — -1

e $\log_{10}\left(\frac{1}{100}\right)$ — -2

f $\log_{10}\left(\frac{1}{100000}\right)$ — -5

g $\log_2 10 - \log_2 5$ — 1

h $\log_3 30 - \log_3 10$ — 1

i $\log_4 128 - \log_4 2$ — 3

j $\log_4 8 + \log_4 2$ — 2

k $\log_8 16 + \log_8 4$ — 2

l $\log_{10} 50 + \log_{10} 2$ — 2

10 Simplify to a single logarithm.

a $2\log_3 2 + \log_3 5$ — $\log_3 20$

b $4\log_{10} 2 + \log_{10} 3$ — $\log_{10} 48$

c $3\log_{10} 2 - \log_{10} 4$ — $\log_{10} 2$

d $5\log_7 2 - \log_7 16$ — $\log_7 2$

e $\frac{1}{2}\log_3 4 + 2\log_3 2$ — $\log_3 8$

f $\log_5 3 - \frac{1}{2}\log_5 9$ — 0

g $\frac{1}{3}\log_2 27 - \frac{1}{3}\log_2 64$ — $\log_2 \frac{3}{4}$

h $\frac{1}{4}\log_5 16 + \frac{1}{5}\log_5 243$ — $\log_5 6$

11 Simplify:

a $\log_2\sqrt{8}$

$= \log_2 8^{\frac{1}{2}}$

$= \log_2 2^{\frac{3}{2}}$

$=$

$\frac{3}{2}$

b $\log_2 \sqrt[3]{32}$

$\frac{5}{2}$

c $\log_2 \sqrt[4]{16}$

$\frac{4}{3}$

d $\log_{10}\sqrt{1000}$

$\frac{3}{2}$

e $\log_7 \sqrt[3]{7}$

$\frac{1}{3}$

f $\log_5 \sqrt[5]{625}$

$\frac{4}{5}$

12 Simplify:

a $\log_a a^2$

2

b $5\log_a a^3$

15

c $\log_a \frac{1}{a}$

-1

d $12\log_a \sqrt{a}$

6

13 Write in terms of $\log_a x$

a $\log_a x^3$

$3\log_a x$

b $\log_a \frac{1}{x}$

$-\log_a x$

c $\log_a \sqrt{x}$

$\frac{1}{2}\log_a x$

d $\log_a \frac{1}{x^2}$

$-2\log_a x$

e $\log_a x^3 - \log_a x^5$

$-2\log_a x$

f $\log_a x^4 + \log_a \frac{1}{x^2}$

$2\log_a x$

g $2\log_a a^4 - \log_a x^8$

$8 - 8\log_a x$

h $\log_a \frac{1}{\sqrt{x}} + 3\log_a \sqrt{x}$

$-\frac{1}{2}\log_a x$

14 If $x = \log_a 2, y = \log_a 3$ and $z = \log_a 5$, write in terms of x, y, z:

a $\log_a 64$

$6x$

b $\log_a \frac{1}{30}$

$-x - y - z$

c $\log_a 27\,a^5$

$3y + 5$

d $\log_a \frac{100}{a}$

$2x + 2z - 1$

e $\log_a 1.5$

$y - x$

f $\log_a \frac{18}{25a}$

$x + 2y - 2z - 1$

g $\log_a 0.04$

$-2z$

h $\log_a \frac{8}{15a^2}$

$3x - y - z - 2$

15 Give exact values of:

a $\log_{25} 45 - \log_{25} 9$

$\frac{1}{2}$

b $\log_{81} 12 - \log_{81} 36$

$-\frac{1}{4}$

c $\log_8 3 - \log_8 96$

$-\frac{5}{3}$

d $\log_{\frac{1}{32}} 5 - \log_{\frac{1}{32}} 10$

$\frac{1}{5}$

16 Simplify:

a $5^{-\log_5 2}$

$\frac{1}{2}$

b $12^{2\log_{12} 7}$

49

c $2^{\log_2 3 + \log_2 5}$

15

d $a^{n \log_a x}$

x^n

e $7^{-\log_7 x}$

$\frac{1}{x}$

f $5^{x+\log_5 x}$

$x \times 5^x$

g $2^{x \log_2 x}$

x^x

h $3^{\frac{\log_3 x}{x}}$

$x^{\frac{1}{x}}$

17 Rewrite in index form by making both sides a single logarithm then exponentiating both sides.

a $\log_a(x+y) = \log_a c + \log_a d$

$x + y = cd$

b $\log_{10} x = 3 + \log_{10} y$

$x = 1000y$

c $\log_3 x = 4 \log_3 y$

$x = y^4$

d $2\log_2 x + 3\log_2 y - 4\log_2 z = 0$

$x^2y^3 = z^4$

e $x \log_a 2 = \log_a y$

$2^x = y$

f $\log_a x - \log_a y = n \log_a z$

$x = yz^n$

g $\frac{1}{2}\log_2 x = \frac{1}{3}\log_2 y - 1$

$64x^3 = y^2$

h $2\log_3(2x+1) = 3\log_3(2y-1)$

$(2x+1)^2 = (2y-1)^3$

18 HSC Sample Question Band 4

Solve $100^{x-2} = 1000^{x}$

$$(10^2)^{x-2} = (10^3)^x$$
$$10^{2(x-2)} = 10^{3x}$$
$$2(x-2) = 3x$$
$$2x - 4 = 3x$$
$$x = -4$$

19 HSC Sample Question Band 4

Solve $2^{3x-3} = 8^{2-x}$

$$2^{3x-3} = (2^3)^{2-x}$$
$$2^{3x-3} = 2^{3(2-x)}$$
$$3x - 3 = 6 - 3x$$
$$6x = 9$$
$$x = \frac{3}{2}$$

20 2017 HSC Band 4

Solve $2\log_3(5) - \log_3(2) + \log_3(x) = 2$

$$log_3\left(\frac{25}{2}\right) + log_3(x) = 2$$
$$log_3\left(\frac{25x}{2}\right) = 2$$
$$\frac{25x}{2} = 3^2$$
$$25x = 18$$
$$x = \frac{18}{25}$$

21 HSC Sample Question Band 4

Solve $\log_2(6-x) - \log_2(4-x) = 2$, when $x < 4$

$$log_2\left(\frac{6-x}{4-x}\right) = 2$$
$$\frac{6-x}{4-x} = 4$$
$$6 - x = 4(4-x)$$
$$6 - x = 16 - 4x$$
$$3x = 10$$
$$x = \frac{10}{3}$$

Solving Equations and Change of Base Law

ⓘ Change of Base Law

Calculators with the general log button $\log_{\blacksquare}\square$ can evaluate any logarithm. E.g. $\log_2 8 = 3$.

Calculators with only the $\log$ button can only evaluate log base 10
The 10 is hidden as most engineering applications use base 10: $\therefore \log x = \log_{10} x$

To evaluate logarithms on a basic calculator, we use the **change of base law:**

$$\log_a b = \frac{\log_c b}{\log_c a}$$

This lets you convert any logarithm to a base your calculator can handle (usually base 10).

⚠ Many questions involving simplifying logarithms requires the change of base law.

🔍 **Rewrite** logarithms using the change of base law.

$\log_2 8 = \frac{\log_{10} 8}{\log_{10} 2}$	$\log_2 8 = \frac{\log_{17} 8}{\log_{17} 2}$	$\log_5 9 = \frac{\log_{17} 9}{\log_{17} 5}$

Rewrite these logarithms with the required base using the change of base law.

a $\log_3 81 = \frac{\log_{10}}{\log_{10}}$	**b** $\log_3 81 = \frac{\log_5}{\log_5}$	**c** $\log_7 22 = \frac{\log_e}{\log_e}$
d $\log_6 36$, base 10	**e** $\log_8 21$, base 15	**f** $\log_2 5$, base a

Solving Equations of the form $a^x = b$ using Logarithms

1. Rewrite to make x the subject using the rule:

$$a^x = b$$
$$x = \log_a b$$

2. Evaluate directly using $\log_{\square}$ or using the change of base law and $\log$

Example Solve equations of the form $a^x = b$ using logarithms.

Solve $2^x = 7$ to 3 d.p.

Method 1 using calculator with $\log$

$$2^x = 7$$
$$x = \log_2 7$$
$$x = \frac{\log_{10} 7}{\log_{10} 2}$$
$$= 2.807$$

Method 2 using calculator with $\log_{\square}$

$$2^x = 7$$
$$x = \log_2 7$$
$$= 2.807$$

Practice

Solve to 3 d.p.

a $3^x = 15$

2.465

b $9^x = 3$

$x = 0.500$

c $3 = 27^x$

0.333

d $4^x = -2$

No solution

e $50 \times 1.1^x = 100$

7.273

f $4^{x+1} = \frac{1}{8\sqrt{2}}$

-2.750

1 Solve the following and round to three decimal places where necessary.

a	$2^x = 6$	**b**	$3^x = 8$	**c**	$5^x = 7$
	2.585		1.893		1.209
d	$11^x = 15$	**e**	$1.8^x = 2.5$	**f**	$0.9^x = 0.5$
	1.129		1.559		6.579
g	$10 \times 2^x = 100$	**h**	$7 \times 3^x = 28$	**i**	$130 \times 7^x = 260$
	3.322		1.262		0.356
j	$4 \times 1.5^x = 20$	**k**	$100 \times 0.8^x = 50$	**l**	$30 \times 0.7^x = 20$
	3.969		3.106		1.137

2 Solve these equations. Answer to 3 d.p. where appropriate.

a	$5^x = 125$	**h**	$4^x = 512$	**o**	$3^x = 13$
	3		4.5		2.335
b	$5^x = 625$	**i**	$2^x = 512$	**p**	$3^x = 169$
	4		9		4.669
c	$5^x = 243$	**j**	$512 = 2^x$	**q**	$3^x = 13^2$
	3.413		9		4.669
d	$3^x = 243$	**k**	$512 = x^9$	**r**	$3^x = 13^3$
	5		2		7.004
e	$3^x = 2187$	**l**	$2^x = \frac{1}{512}$	**s**	$13^x = 169$
	7		−9		2
f	$3^x = 512$	**m**	$4^x = \frac{1}{512}$	**t**	$13^x = \frac{1}{169}$
	5.678		−4.5		−2
g	$8^x = 512$	**n**	$8^x = \frac{1}{512}$	**u**	$3^{2x+1} = 169$
	3		−3		1.835

3

a Use the substitution $u = 2^x$ to solve $4^x - 9 \times 2^x + 14 = 0$.

$x = 1$ or $x = \log_2 7$

b Use the substitution $u = 3^x$ to solve $3^{2x} - 8 \times 3^x - 9 = 0$.

$x = 2$, as $(3^x = -1)$ has no solutions

4 Use similar substitutions to solve:

a $25^x - 26 \times 5^x + 25 = 0$

$x = 2$ or 0

b $9^x - 5 \times 3^x + 4 = 0$

$x = 0$ or $\log_3 4$

c $3^{2x} - 3^x - 20 = 0$

$x = \log_3 5$

p $7^{2x} + 7^x + 1 = 0$

No solutions

q $2^{2x} - 64 = 0$

$x = 3$

r $2^x - 3 \times 2^{\frac{x}{2}} + 2 = 0$

$x = 2$ or 0

5 Give exact solutions to these inequations. Do not use a calculator.
Remember that the inside of a logarithm cannot be zero or negative.

a $\log_2 x < 3$

$0 < x < 8$

b $\log_2 x \geq 3$

$x \geq 8$

c $\log_{10} x > 3$

$x > 1000$

d $\log_6 x < 1$

$0 < x < 6$

6 Rewrite each inequation in terms of logarithms, with x as the subject. Then solve it, giving your answer correct to three significant figures.

a $2^x > 12$

$x > 3.58$

b $2^x < 100$

$x < 6.64$

c $5^x < 100$

$x < 2.86$

d $3^x > 0.007$

$x > -4.52$

7 By using the change of base law

$$\log_a b = \frac{\log_n b}{\log_n a}$$

Simplify the following **without** a calculator by changing the base to the specified value.

a $\log_8 32$ (base 2)

$\frac{\log_2 32}{\log_2 8} = \frac{5}{3}$

b $\log_{0.25} 32$ (base 2)

$\frac{\log_2 32}{\log_2 \frac{1}{4}} = \frac{5}{-2}$

c $\log_{\frac{1}{9}} 27$ (base 3)

$\frac{\log_3 27}{\log_3 \frac{1}{9}} = \frac{3}{-2}$

d $\log_{25} 125$ (base 5)

$\frac{\log_5 125}{\log_5 25} = \frac{3}{2}$

8

a Using the change of base law, show that $\log_a b = \frac{1}{\log_b a}$

$\log_a b = \frac{\log_b b}{\log_b a} = \frac{1}{\log_b a}$

b Hence find the value of $\log_{\sqrt{27}}(3)$.

$log_{\sqrt{27}}(3) = \frac{1}{\log_3 \sqrt{27}} = \frac{1}{\left(\frac{3}{2}\right)} = \frac{2}{3}$

9 Consider $\log_5 y = 3 - 2\log_5 x$.

By changing the subject to y, show that $y = \frac{5^3}{x^2}$

$\log_5 y = 3 - \log_5 x^2$

$\log_5 y + \log_5 x^2 = 3$

$\log_5(yx^2) = 3$

$yx^2 = 5^3$

$y = \frac{5^3}{x^2}$

10 Consider $p = \frac{\log_{10} a}{\log_{10} x}$. Show that $x^p = a$

$p = \log_x a$

$x^p = a$

11 Prove that $\log_{xy}(a) = \frac{\log_x(a)}{1+\log_x(y)}$

$$RHS = \frac{\log_x a}{\log_x x + \log_x y}$$
$$= \frac{\log_x a}{\log_x xy}$$
$$= \log_{xy} a$$
$$= LHS$$

12 Prove that $\log_{a^x}(b) = \frac{\log_a b}{x}$

$$RHS = \frac{\log_a b}{\log_a a^x}$$
$$= \log_{a^x} b$$
$$= LHS$$

13 Solve $\log_8 x + \log_4 x = 5$

$$\frac{\log_2 x}{\log_2 8} + \frac{\log_2 x}{\log_2 4} = 5$$
$$\frac{\log_2 x}{3} + \frac{\log_2 x}{2} = 5$$
$$\frac{5\log_2 x}{6} = 5$$
$$\log_2 x = 6$$
$$x = 2^6$$

14 By writing in index form, show the following is true.

a $\log_8 64 = 2$

$64 = 8^2$

b $\log_{64} 8 = \frac{1}{2}$

$8 = 64^{\frac{1}{2}}$

c $\log_8 \frac{1}{64} = -2$

$\frac{1}{64} = 8^{-2}$

d $\log_{64} \frac{1}{8} = -\frac{1}{2}$

$\frac{1}{8} = 64^{-\frac{1}{2}}$

Hence evaluate the following:

e $\log_2 8$

$x = \log_2 8$
$2^x = 8$
$x = 3$

f $\log_8 2$

$x = \log_8 2$
$8^x = 2$
$x = \frac{1}{3}$

g $\log_2 \frac{1}{8}$

$x = \log_2 \frac{1}{8}$
$2^x = \frac{1}{8}$
$x = -3$

h $\log_8 \frac{1}{2}$

$x = \log_8 \frac{1}{2}$
$8^x = \frac{1}{2}$
$x = -\frac{1}{3}$

15 Simplify the following expressions:

a $\log_a a^n$

n

b $\log_{(a^n)} a$

$\frac{1}{n}$

c $\log_a \frac{1}{a^n}$

$-n$

d $\log_{a^n} \frac{1}{a}$

$-\frac{1}{n}$

16 2013 HSC Advanced Band 4

What is the solution of $5^x = 4$?

(A) $x = \dfrac{\log_e 4}{5}$

(B) $x = \dfrac{4}{\log_e 5}$

(C) $x = \dfrac{\log_e 4}{\log_e 5}$

(D) $x = \log_e\left(\dfrac{4}{5}\right)$

$x = \log_5 4$

$= \dfrac{\log_e 4}{\log_e 5}$

C

17 2019 HSC Advanced Band 4

Which of the following is equal to $\dfrac{\log_2 9}{\log_2 3}$?

(A) 2

(B) 3

(C) $\log_2 3$

(D) $\log_2 6$

$\log_3 9 = 2$

A

18 2019 VCE Band 5

Solve $\log_3(t) - \log_3(t^2 - 4) = -1$

$\log_3\left(\dfrac{t}{t^2-4}\right) = -1$

$\dfrac{t}{t^2-4} = 3^{-1}$

$\dfrac{t}{t^2-4} = \dfrac{1}{3}$

$3t = t^2 - 4$

$t^2 - 3t - 4 = 0$

$(t-3)(t+1) = 0$

$t = 3$ or -1

But $t \neq -1$ as $\log_3 t$ would be undefined

$\therefore t = 3$

19 2020 VCE Band 5

Solve the equation $2\log_2(x+5) - \log_2(x+9) = 1$

$\log_2(x+5)^2 - \log_2(x+9) = 1$

$\log_2\left(\dfrac{(x+5)^2}{x+9}\right) = 1$

$\dfrac{(x+5)^2}{x+9} = 2$

$(x+5)^2 = 2(x+9)$

$x^2 + 10x + 25 = 2x + 18$

$x^2 + 8x + 7 = 0$

$(x+7)(x+1) = 0$

$x = -7, -1$

But $x \neq -7$ as $\log_2(x+5)$ would be undefined, $\therefore x = -1$

20 Use the given fact to evaluate the given expression.

a $\log_5 125 = 3$

$\log_{125} 5 = ?$

$\frac{1}{3}$

b $\log_{100} 10 = \frac{1}{2}$

$\log_{10} 100 = ?$

2

c $\log_4 \frac{1}{64} = -3$

$\log_{64} \frac{1}{4} = ?$

$-\frac{1}{3}$

d $\log_{36} 6 = \frac{1}{2}$

$\log_{36} \frac{1}{6} = ?$

$-\frac{1}{2}$

21 Fill in the blanks.

a $\log_4 16 = \log_2 \square = \log_\square 9$

4, 3

b $\log_2 16 = \log_{10} \square = \log_\square 81$

100, 3

c $\log_{16} 4 = \log_{100} \square = \log_\square 2$

10, 9

d $\log_2 \frac{1}{2} = \log_3 \square = \log_\square \frac{1}{5}$

$\frac{1}{3}$, 5

e $\log_4 8 = \log_{16} \square = \log_\square 27$

64, 9

22 An advanced technique for solving exponential equations is to take the logarithm of both sides. This is useful when the bases are different. Answer to 4 s.f.

Example

Worked example	Your Turn
$2^{3x} = 3^{x+1}$ $\log_{10} 2^{3x} = \log_{10} 3^{x+1}$ $3x \log_{10} 2 = (x+1) \log_{10} 3$ $3x \log_{10} 2 = x \log_{10} 3 + \log_{10} 3$ $3x \log_{10} 2 - x \log_{10} 3 = \log_{10} 3$ $x(3 \log_{10} 2 - \log_{10} 3) = \log_{10} 3$ $x = \frac{\log_{10} 3}{3 \log_{10} 2 - \log_{10} 3}$ ≈ 1.120	$5^{2x} = 6^{x+1}$ 1.256

23 Solve $7^{1-x} = 6 \times 5^{x-3}$.

1.401

Exponential Modelling

ⓘ Solving Problems involving Exponential Models

When solving exponential equations, use logarithms to find the missing index.

1. Write the formula.
2. Substitute given values.
3. Isolate exponential term.
4. Solve using logarithms.

Example Solve exponential modelling problems using logarithms.

An investment of \$10 000 grows by 5% p.a according to the compound interest formula:

$$FV = 10000(1.05)^n$$

Where n is the number of years. Find the time required for the investment to grow to \$32000.

1. Formula $FV = 10000(1.05)^n$
2. Substitute $32000 = 10000(1.05)^n$
3. Isolate $3.2 = 1.05^n$
4. Solve $\log_{1.05} 3.2 = n$

$n \approx 23.84$ years

Practice

An investment of \$20 000 grows by 6% p.a according to the compound interest formula:

$$FV = 20000(1.06)^n$$

Where n is the number of years. Find the time required for the investment to grow to \$50000. Round to 2 decimal places.

15.73 years

A population of mosquitoes is given by $P = 8^t$ where t is measured in days.
Find the number of days, correct to 2 decimal places, required for the population to reach 200.

2.55 days

1 The number of plankton in a tank is doubling every 5 days, given by the formula:

$$P = 15000 \times 2^{\frac{t}{5}}$$

Where P is population (number of plankton) after t days from commencement of study.

a What was the initial population when the study commenced?

$t = 0, P = 15000 \times 2^{\frac{0}{5}} = 15000$

b What was the population after 15 days?

$P = 15000 \times 2^{\frac{15}{5}} = 15000 \times 8 = 120000$

c At what time will the population reach 1 920 000?

$1920000 = 15000 \times 2^{\frac{t}{5}}$

$128 = 2^{\frac{t}{5}}$

$2^7 = 2^{\frac{t}{5}}$

$\frac{t}{5} = 7$

$t = 35$ days

2 The temperature in a tank of water is decreasing according to the formula

$$T = 100 \times (0.87)^h$$

Where T is temperature in degrees Celsius after h minutes.

a What was the initial temperature?

100° C

b How long would it take for the temperature to drop to $70°C$?

$70 = 100 \times (0.87)^h$

$0.7 = (0.87)^h$

$h = \log_{0.87} 0.7 = 2.56$ minutes

3 The mass, in grams, of a decaying substance is given by

$$M = 18 \times (1.75)^{-\frac{3t}{5}}$$

Where t is measured in days.

a What was the initial mass?

18 g

b What is the half-life? (How long will it take to decay to half its current mass?)

$9 = 18 \times (1.75)^{-\frac{3t}{5}}$

$0.5 = (1.75)^{-\frac{3t}{5}}$

$-\frac{3t}{5} = \log_{1.75} 0.5$

$t = \frac{5 \log_{1.75} 0.5}{-3} = 2.06$ days

4 A bacterial population starts with 1000 bacteria and doubles every half hour.

a How many bacteria are there after 1 hour?

4000

b How many bacteria are there after 2 hours?

16000

c How many bacteria are there after 3 hours?

64000

d Write down an equation for the population of the bacteria after t hours.

1000×2^{2t}

e How long will it take for the population to exceed 10 million? Answer to the nearest minute.

$1000 \times 2^{2t} > 10\,000\,000$

$2^{2t} > 10\,000$

$2t > \log_2 10\,000$

$t > 6.64385 \ldots$

6 hours 39 min

5 The value of a bank balance increases by 10% per year. The initial amount is \$2000.

a Write the equation in the form $FV = PV(1 + r)^n$.

$FV = 2000(1.1)^n$

b Find the time, correct to two decimals, when the balance is double the original amount.

$4000 = 2000(1.1)^n$

$2 = 1.1^n$

$n = \log_{1.1} 2$

$= 7.27$ years

6 The value of a car is expected to reduce by 8% per year. The original cost is \$300 000.

a Write the equation in the form $S = V_0(1 - r)^n$.

$S = 300\,000(0.92)^n$

b Find the time it takes for the car's value to be half the original value.
Round your answer to 1 decimal place.

$150000 = 300\,000(0.92)^n$

$\frac{1}{2} = 0.92^n$

$n = \log_{0.92} \frac{1}{2}$

$= 8.3$ years

7 A container of hot water is placed in a refrigerator running at $0°C$. The water is known to cool according to Newton's Law of Cooling; with a given starting temperature of 96°C its temperature halves every 20 minutes. The formula is given by

$$T = 96 \times \left(\frac{1}{2}\right)^{3n}, \text{where } n \text{ is in hours}$$

a What is the temperature after 2 hours?

1.5° C

b How long does it take, correct to the nearest minute, for the temperature to fall to 1°C?

132 minutes

8 The population of a country is doubling every 30 years and was 3 000 000 at the last census.

a Show, by establishing a pattern, why the population P after n years is $P = 3\,000\,000 \times 2^{\frac{n}{30}}$

$n = 0: P = 3{,}000{,}000 \times 2^0$

$n = 30: P = 3{,}000{,}000 \times 2^1$

$n = 60: P = 3{,}000{,}000 \times 2^2$

$\therefore P = 3{,}000{,}000 \times 2^{\frac{n}{30}}$

b Find when the population will be 1 billion, to 1 d.p.

251.4 years

9 The half-life of a substance is the time it takes for the substance to reduce to half its original mass. Find the half-life of a radioactive material if its mass reduces by 1% per year.

Let M be original mass.

$\frac{M}{2} = M \times (0.99)^t$

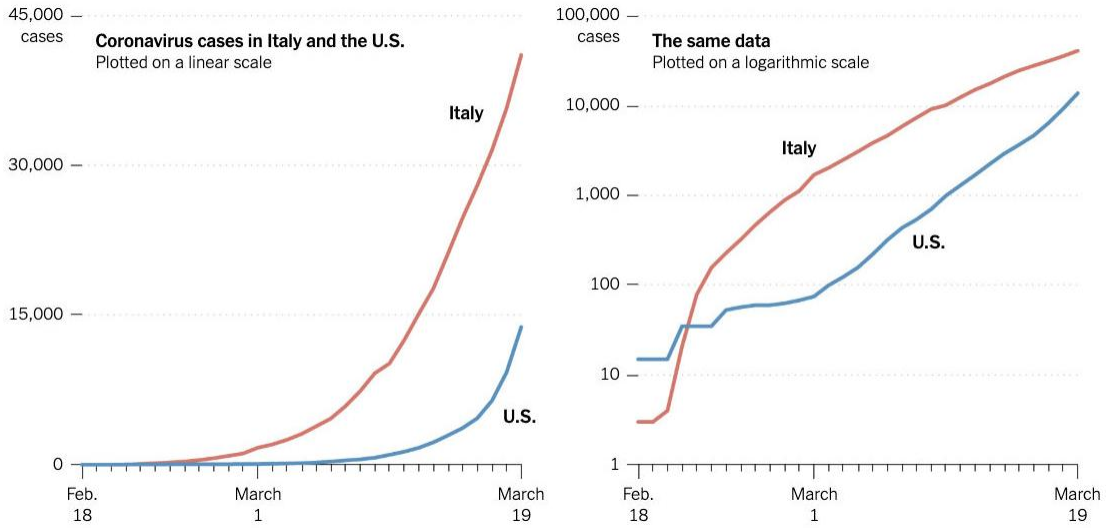
$0.5 = (0.99)^t$

$t = \log_{0.99} 0.5 = 69.0 \text{ years}$

Logarithmic Scales

Interpret a logarithmic scale.

Total COVID-19 cases in Italy and the US in early 2020:



a What scale is the second graph using? How can you tell?

b What happens to an exponential relationship when you use a logarithmic scale?
Hint: It's difficult to tell using real-world data that is constantly changing.
What happens to $y = 2^x$ when you do $y = \log_{10}(2^x)$?

c Using the logarithmic scale, what can you say about Italy's case numbers?
Is it slowing down or accelerating?

d Using the logarithmic scale, what can you say about USA's case numbers?
Is it slowing down or accelerating?

e Why might scientists prefer to use logarithmic scales to track exponential growth scenarios, such as infectious diseases?

Example Interpret logarithmic scales.

A Richter scale is used to measure Earthquake intensity.

$$M = \log_{10} A$$

Where M is the magnitude of the earthquake and A is the amplitude of the ground movements

Rewrite this formula with A as the subject.

$10^M = A$

How many times more powerful is a magnitude 6 earthquake compared to magnitude 3?

Magnitude 6: $A = 10^6$

Magnitude 3: $A = 10^3$

$\frac{10^6}{10^3} = 1000$ times as powerful

Practice

pH is used to measure the hydrogen ion concentration in a substance, hence telling us how acidic or basic it is.

$$p = -\log_{10} H$$

Where p is the pH of the substance and H is the hydrogen ion concentration.

Rewrite this formula with H as the subject.

$10^{-p} = H$

Lemonade has a pH of 3 and sparkling water a pH of 5.

How many times more acidic is lemonade compared to sparking water?

$\frac{10^{-3}}{10^{-5}} = 100$ times as acidic

1 Earthquake intensity is measured using the Richter scale

$$M = \log_{10} A$$

M is the magnitude of an earthquake.

A is the amplitude of ground movements.

a Rewrite the formula with A as the subject.

How many times more powerful are these earthquakes compared to each other?
i.e. find the ratio of the amplitudes.

Answer to 2 s.f.

b Magnitude 2 (smallest humans can feel)
Magnitude 6 (major damage)

$\frac{10^6}{10^2} = 10\,000$ times as powerful

c 3.5 and 6?

$\frac{10^6}{10^{3.5}} = 320$ times as powerful

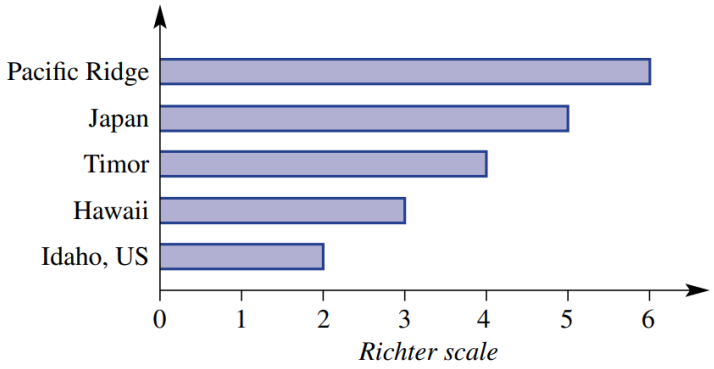
d 3.2 and 8.6?

250000 times as powerful

e 2.8 and 3.6?

6.3 times as powerful

2 This chart shows the Richter scale measurements rounded to the nearest integer for five different earthquakes recorded in 2022.



a What is the difference between the magnitude of the earthquakes for Pacific Ridge compared to Timor?

2

b Therefore, explain why the earthquake in Pacific ridge was 100 times more powerful compared to Timor.

$10^2 = 100$

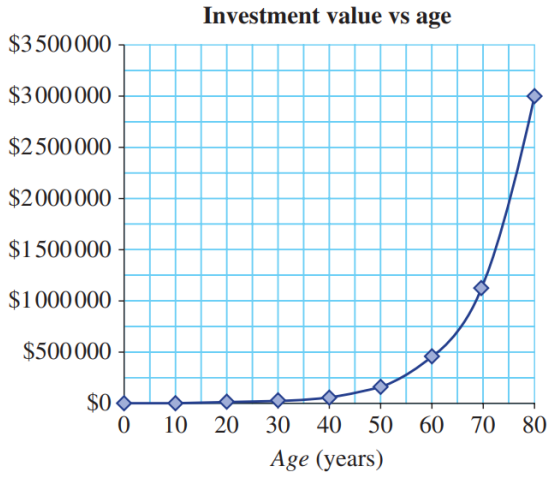
c What is the difference between the magnitude of the earthquakes for Japan compared to Idaho?

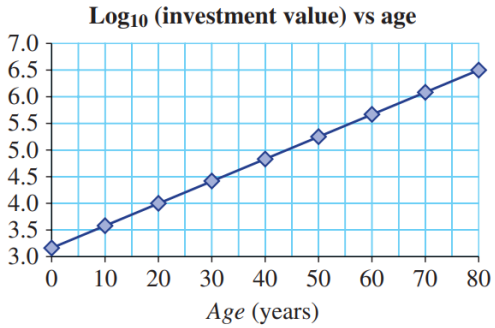
3

d Therefore, how many times more powerful was the earthquake in Japan compared Idaho?

1000

3 The two graphs shows \$1500 earning compound interest over 80 years.





a What investment value does 3.6 correspond to on the log scale?

$10^{3.6} = \$3981$

b Which graph can be used to find the investment value at age 20 years and what is this value?

The log graph. $10^4 = 10000$

c By what factor has the investment value increased from age 20 to age 30?

Looking at the log graph, the difference between 30 and 40 is around 0.4

$10^{0.4} \approx 2.5$

2.5 times

d By what factor has the investment value increased when the log scale increases by 1 unit?

10 times

e If a graph using a log scale is linear, what form is the graph of the original data?

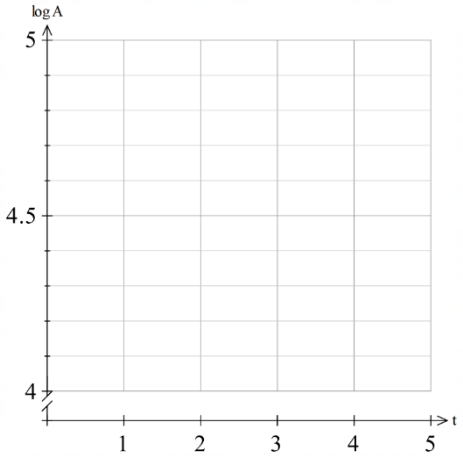
Exponential

4 The following table includes data about the value of an investment \$A over t years.

a Calculate the value of $\log_{10} A$ for each of the values of A on the table.

t	0	1	2	3	4	5
A	10 000	12 000	14 400	17 280	20 736	24 883.2
$\log_{10} A$						

b Plot $(t, \log_{10} A)$ on this graph.



c What does this say about the type of relationship between t and and A?

Since the log graph is linear, the original must be exponential.

5 The pH of a liquid is traditionally defined on a logarithmic scale as $\text{pH} = -\log_{10}(x)$, where x is the hydrogen ion concentration in moles per litre (mol/L). The greater the concentration of hydrogen ions, the more acidic it is.

a Rewrite the formula with x as the subject.

$x = 10^{-\text{pH}}$

b Pure water has a pH of about 7. What is its hydrogen ion concentration?

10^{-7} mol/L

c Lemon juice typically has a pH of about 2.0. What is its hydrogen ion concentration? How many times more acidic is it than pure water?

0.01 mol/L

$\frac{0.01}{1 \times 10^{-7}}$ = 100,000 times more acidic than pure water

d Sea water typically has a pH of about 8. What is its hydrogen ion concentration?

How many times more alkaline (meaning 'less acidic') is it than pure water?

10^{-8} mol/L

10 times less acidic (more alkaline) than pure water

6 The human ear is very sensitive. Sound intensity is measured using decibels (dB).

$$S = 10 \log_{10} \frac{I}{I_0}$$

Where S is the sound intensity in decibels,

$\frac{I}{I_0}$ is how many times louder a sound is compared to the threshold of human hearing.

How many times louder than the threshold for human hearing is the sound of the following? (i.e., Find $\frac{I}{I_0}$)

a Normal conversation
60 dB

$60 = 10 \log_{10} \frac{I}{I_0}$

$6 = \log_{10} \frac{I}{I_0}$

$\frac{I}{I^0} = 10^6 = 1{,}000{,}000$

b Whispering
20 dB

100

c How many times louder is 60 dB vs 20 dB?

10 000 times louder

7 Using a log base 10 scale, the magnitude of the volume of water in Lake Victoria is 8 and the magnitude of the volume of water in Lake Anne is 5.
There is 500 000 ML of water in Lake Anne.



That means Lake Victoria has 800 000 ML of water.

Explain why this is incorrect and state the correct volume of Lake Victoria.

Using a base 10 scale, a difference of 3 means a difference of 10^3 in the original scale.

Lake Victoria has 500 000 000 ML of water.

The Logarithmic Curve

Investigation Graph logarithmic curves using a table of values.

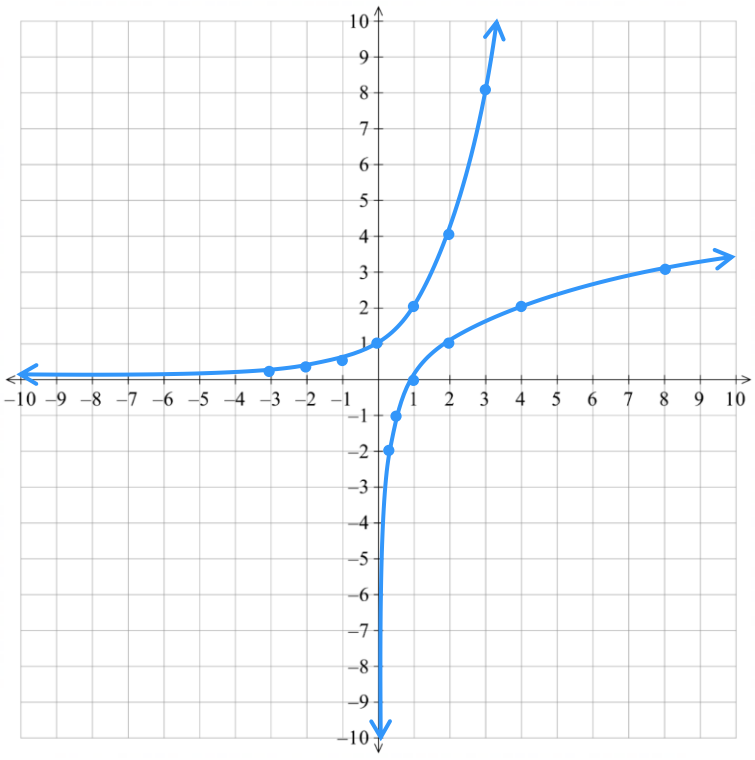
Sketch the graphs of $y = 2^x$ and $y = \log_2 x$ on the same axis.

$y = 2^x$

x	-3	-2	-1	0	1	2	3
y							

$y = \log_2 x$

x	-1	0	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8	16
y									



Why can't the inside of a logarithm be zero or negative?

The definition of a logarithm says that the inside of a logarithm must be .....

Consider $x = \log_2 -1$, changing to index form: ..... = ..., which is impossible.

What do you notice about the graphs of $y = 2^x$ and $y = \log_2 x$?

They are ..... of each other across the line .....

Where are the asymptotes for $y = 2^x$ and $y = \log_2 x$? .....

Investigation Logarithmic Curve Transformations.

Using Desmos:

1 Sketch

$y = a^x$

$y = \log_a x$

Use the slider to change the value of a.

Are the graphs always reflections of each other across $y = x$?

2 Sketch

$y = \log_2 x$

$y = \log_3 x$

$y = \log_{1.1} x$

$y = \log_{0.9} x$

$y = \log_{\frac{1}{2}} x$

When $a > 1$, the graph always has a .....

gradient; it is always i.....

When $0 < a < 1$, the graph always has a .....

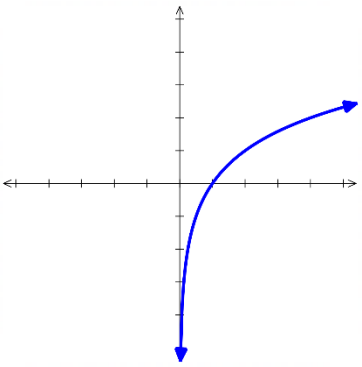
gradient, it is always d.....

Logarithmic Curve Transformations

For a logarithmic curve $y = log_a(x)$:

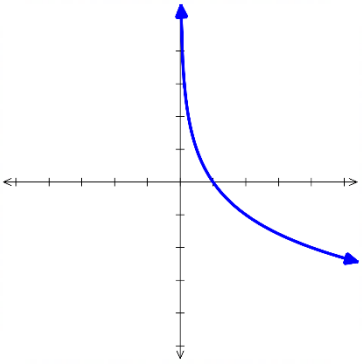
$a > 0$, the curve is increasing.

$y = \log_2 x$



$0 < a < 1$, the curve is decreasing

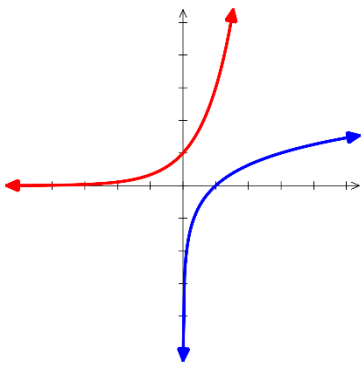
$y = \log_{0.5} x$



$y = \log_a x$ and $y = a^x$ are reflections of each other across the mirror line $y = x$

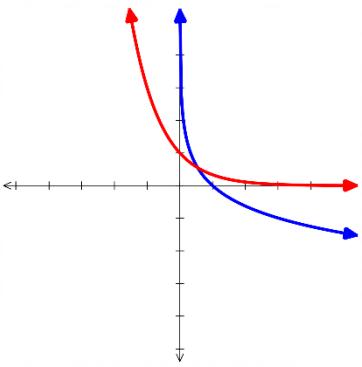
$y = \log_3 x$

$y = 3^x$



$y = \log_{\frac{1}{3}} x$

$y = \left(\frac{1}{3}\right)^x$



ⓘ Natural Domain of a Logarithmic Function

For a logarithm,

$$y = \log_a\big(f(x)\big)$$

The natural domain is $f(x) > 0$.

Identify Domain of a logarithmic function

$\log_3(x-4)$	$\log_{10}(x+4)$	$\log_{10}(x+4)$	$\log_{10}(2x+4)$
$x-4>0$ $x>4$	$x+4>0$ $x>-4$	$x+4>0$ $x>-4$	$2x+4>0$ $x>-2$

Identify the domain of these logarithmic functions.

a $\log_2(x-7)$	**b** $\log_2(x+7)$	**c** $\log_2(3x+7)$
d $\log_2(7-3x)$	**e** $\log_5 x+7$	**f** $\log_4(2x-5)-11$

1 For the function $y = \log_4 x$,

a complete the table:

x	$\frac{1}{16}$	$\frac{1}{4}$	4	16
y				

b hence state whether $y = \log_4 x$ is an increasing or decreasing.

increasing

2 For the function $y = \log_{\frac{1}{2}} x$,

a complete the table:

x	$\frac{1}{2}$	1	2	4
y				

b hence state whether $y = \log_{\frac{1}{2}} x$ is an increasing or decreasing.

decreasing

3 State whether these functions are increasing or decreasing.

a $y = \log_3 x$

increasing

b $y = \log_3(x - 4)$

increasing

c $y = \log_3(5x)$

increasing

d $y = \log_{\frac{1}{3}} x$

decreasing

e $y = \log_{0.3}(x - 4)$

decreasing

f $y = \log_{25}(5x)$

increasing

g $y = \log_{1.01} x$

increasing

h $y = \log_{0.99} x$

decreasing

i $y = \log_3(-x)$

decreasing

4 State the natural domain of these functions.

a	$y = \log_2(x-3)$	**b**	$y = \log_2 x - 3$	**c**	$y = \log_2(x+4)$
	$x > 3$		$x > 0$		$x > -4$
d	$y = \log_2 x + 4$	**e**	$y = \log_2(x-4) + 3$	**f**	$y = \log_5(x-2) + 1$
	$x > 0$		$x > 4$		$x > 2$
g	$y = \log_{10}(4x+1)$	**h**	$y = \log_2(-x)$	**i**	$y = \log_2(5-2x)$
	$x > -\frac{1}{4}$		$x < 0$		$x < \frac{5}{2}$

5 Sketch each set of graphs on Desmos and answer the questions.

$y = \log_2(x+1)$	$y = \log_2(x+7)$	$y = \log_2(x-3)$

a How does the value of b in $\log_a(x-b)$ affect the graph?

Horizontal translation

$y = \log_2(x) + 1$	$y = \log_2(x) + 7$	$y = \log_2(x) - 3$

b How does the value of c in $\log_a(x) + c$ affect the graph?

Vertical translation

6 For the graph of $y = \log_2(x-4) + 3$:

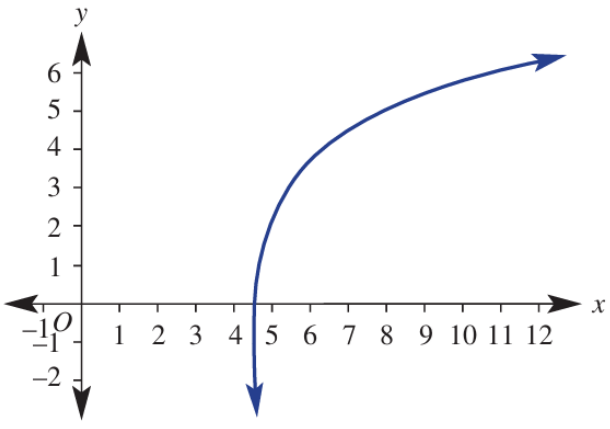
a State the domain.

$x > 4$

b State the equation of the asymptote.

$x = 4$

c Determine the x-intercept by letting $y = 0$.



$(4.125, 0)$

d State the horizontal and vertical translations applied to the graph of $\log_2 x$.

Right by 4, up by 3

7 For the graph of $y = \log_2(x - 3) + 2$:

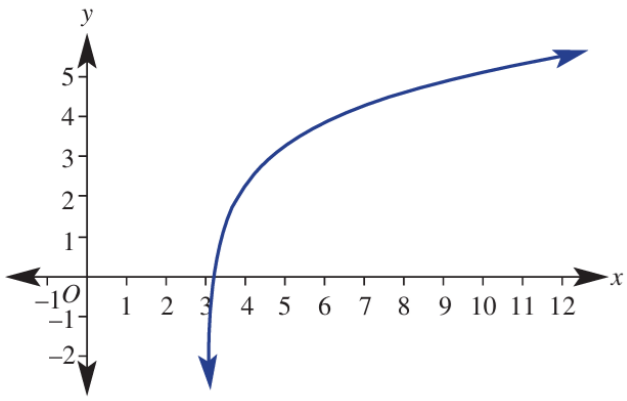
a State the domain.

$x > 3$

b State the equation of the asymptote.

$x = 3$

c Determine the x-intercept by letting $y = 0$.



$(3.25, 0)$

d State the horizontal and vertical translations applied to the graph of $\log_2 x$.

Right by 3, up by 2

8 For the graph of $y = \log_2(x + 4) - 3.5$:

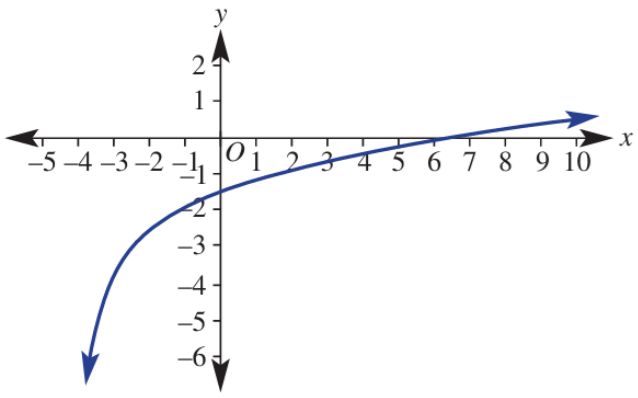
a State the domain.

$x > -4$

b State the equation of the asymptote.

$x = -4$

c Determine the axis intercepts.



$(7.31, 0)$

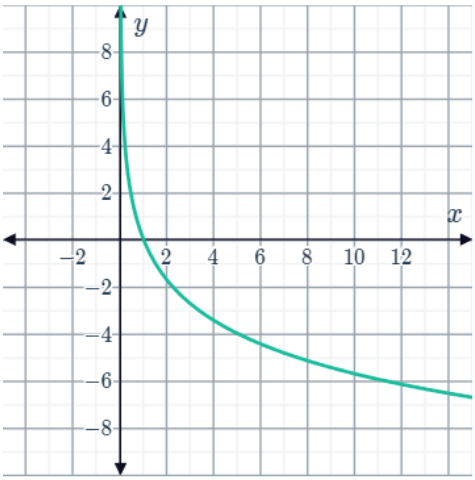
$(0, -1.5)$

d State the horizontal and vertical translations applied to the graph of $\log_2 x$.

left by 4, down by 3.5

9 Consider the curve $y = \log_a x$ below.

Which is a possible value of a: $\frac{2}{3}$ or $\frac{3}{2}$? Explain your answer.

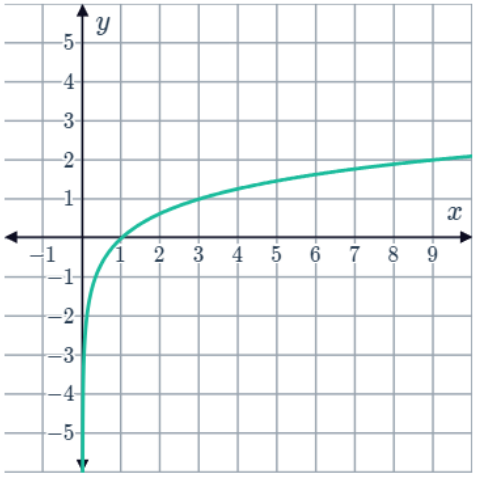


$a = \frac{2}{3}$

The base must be less than one as the graph is decreasing.

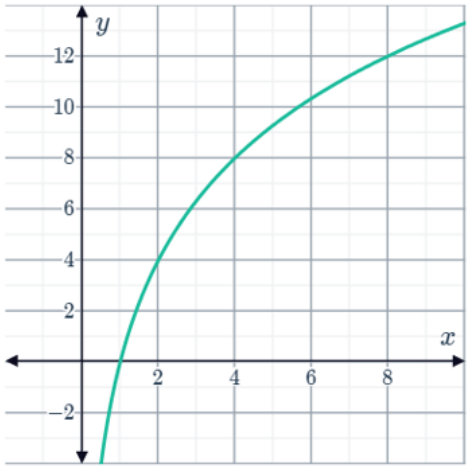
10 Find the equation of the following functions, given it is of the stated form:

a $y = \log_k x$



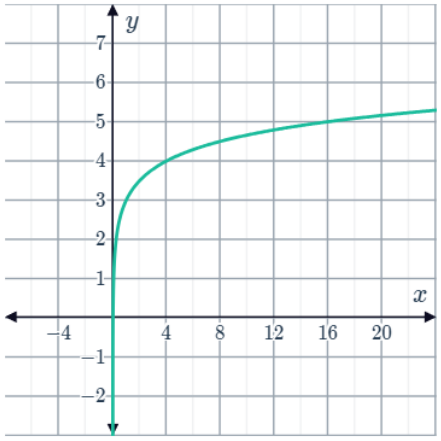
Using (3,1)
$1 = \log_k 3$
Exponentiate both sides base k
$k^1 = 3$
$\therefore k = 3$

b $y = k \log_2 x$



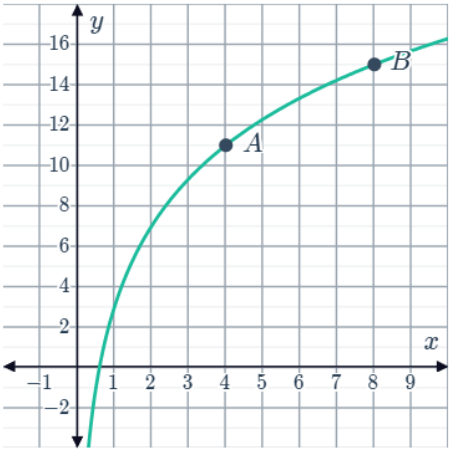
$y = 4 \log_2 x$

c $y = \log_4(x) + c$



$y = \log_4 x + 3$

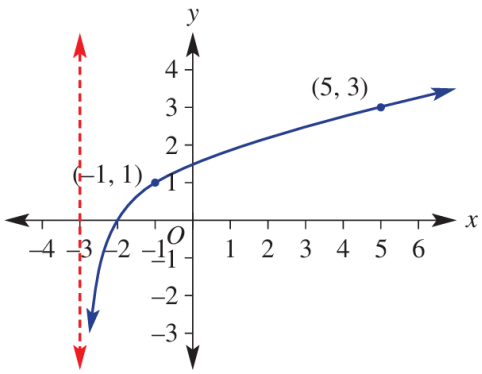
d $y = k \log_2(x) + c$



$y = 4 \log_2 x + 3$

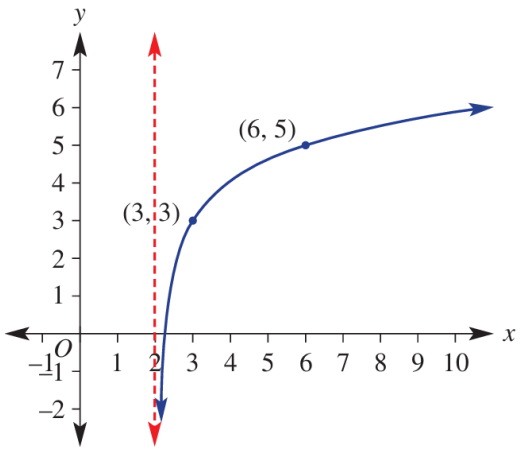
11 Determine the equation for each logarithmic graph in the form $y = \log_a(x - b) + c$.

a



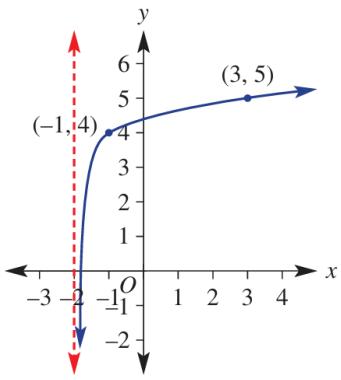
$y = \log_2(x + 3)$

b



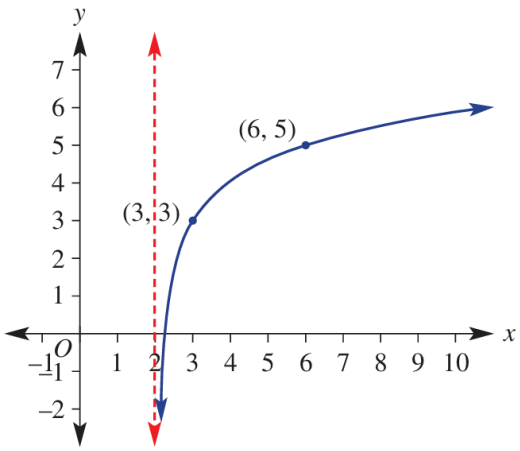
$y = \log_2(x - 2) + 3$

c



$y = \log_5(x + 2) + 4$

d



$y = \log_3(x + 3) - 1$

Logarithmic Inequalities

Review

- [ ] Write exponentials in logarithmic form

a $3^x = 11$

b $(1.2)^x = 2$

- [ ] Solve linear inequalities

a $2x - 5 > 0$

b $-4x + 6 < 11$

- [ ] Identify domain of a logarithm

a $\log_3(x - 5)$

b $\log_{11}(x + 4)$

Reversing the Inequality Sign

Multiplying or dividing both sides of an inequality by a negative flips the inequality sign:

$$5 > 3$$
$$\times -2 \quad \times -2$$
$$-10 < -6$$

But what happens when we take the logarithm of both sides?

Test it out by evaluating both sides.

$$5 > 3$$
$$\log_5 \quad \log_5$$
$$\log_5 5 \ldots\ldots \log_5 3$$

$$5 > 3$$
$$\log_{0.5} \quad \log_{0.5}$$
$$\log_{0.5} 5 \ldots\ldots \log_{0.5} 3$$

When you take the logarithm of a base …… than one, you flip the inequality sign.

The reason for this is that $f(x) = \log_{0.5} x$ is a decreasing function.

The rule is: Applying a strictly decreasing function to both sides of an inequality reverses the direction of the inequality sign.

A strictly decreasing function maps larger inputs to smaller outputs.

This is true of the simple example at the start as well. Instead of thinking of this as multiplying by -2, instead think of it as applying the decreasing function $f(x) = -2x$ to both sides.

$$5 > 3$$
$$f(5) \quad f(3)$$
$$-10 < -6$$

Logarithms and Inequalities

When applying a logarithm to both sides $f(x) = \log_a x$,

Keep the inequality sign the same if $a > 1$.

Reverse the inequality sign if $0 < a < 1$.

Example Solve logarithmic inequalities

$2^x > 3$

$\log_2 \quad \log_2$

$x > \log_2 3$

$0.5^x > 3$

$\log_{0.5} \quad \log_{0.5}$

$x < \log_{0.5} 3$

Guided Practice

Solve the following for x.

a $2^x > 9$

b $5^x < 8$

c $(0.2)^x > 11$

d $\left(\frac{2}{3}\right)^x - 10 < 4$

1 Solve, giving your answer to 3 d.p.

a	$2^x > 5$	**b**	$3^x > 7$	**c**	$5^x < 8$
	$x > 2.322$		$x > 1.771$		$x < 1.292$
d	$10^x < 6$	**e**	$4^x \geq 9$	**f**	$7^x \leq 20$
	$x < 0.778$		$x \geq 1.585$		$x \leq 1.540$
g	$0.5^x > 4$	**h**	$0.5^x < 3$	**i**	$0.25^x > 6$
	$x < -2$		$x > -1.585$		$x < -1.292$
j	$6^x > 11$	**k**	$0.4^x > 11$	**l**	$0.7^x < 12$
	$x > 1.388$		$x < -2.617$		$x > -6.967$
m	$1.05^x > 2$	**n**	$0.95^x > 2$	**o**	$1.2^x \leq 4$
	$x > 14.207$		$x < -13.513$		$x \leq 7.604$

2 Solve:

a	$3(2^x) > 21$	**b**	$4(5^x) < 36$	**c**	$2(0.5)^x > 6$
	$x > 2.807$		$x < 1.365$		$x < -1.585$
d	$5(0.8)^x < 15$	**e**	$7(1.1)^x + 2 > 30$	**f**	$100(0.9)^x < 40$
	$x > -4.923$		$x > 14.545$		$x > 8.697$

3 $1000 is invested at 5% compound interest per year.
After how many whole years will the balance first exceed $2500?

19 years

4 $2000 is invested at 8% per year. After how many whole years will it exceed $20 000?

30 years

5 A car worth $20 000 loses 10% of its value per year.
After how many whole years will its value first fall below $8000?

9 years

6 A machine worth $400 000 loses 15% of its value per year.
After how many whole years will its value first fall below $20 000?

13 years

7 A medicine in the bloodstream loses 20% of its concentration every hour.
If 100 mg is injected at the start, after how long will less than 30 mg remain?
Answer to the nearest minute.

5 hours 24 minutes

8 A disease doubles the number of infected individuals every week.
If 50 people were infected at the start, after how long will 8 billion people be infected?
Give your answer in whole weeks and the nearest day.

27 weeks 2 days

9 The temperature difference between a cup of coffee and its surroundings decreases by 25% every 5 minutes. The coffee is initially 60 degrees hotter than its surroundings; this can be modelled by $D = 60(0.75)^t$ where t is the number of 5 minute periods. After how long will it take for the temperature difference to be less than 5°C? Answer to the nearest whole minute.

44 min

The Natural Logarithm

ⓘ Natural Logarithm

Logarithms of base e are called **natural logarithms.**

$$\log_e x = \ln x$$

e^x and $\log_e x$ are inverses of each other.

Engineers and some calculators use $\log x$ to mean $\log_{10} x$.

However, many mathematicians use $\log x$ to mean $\log_e x$.

Follow these two rules to avoid confusion:

- Always write the base: use $\log_{10} x$ or $\log_e x$, never just "$\log x$"
- For the natural logarithm, always write either $\ln x$ or $\log_e x$

ⓘ Evaluating Natural Logarithms

All log laws apply.

$\ln 1 = 0$	$\ln e^x = x$
$\ln e = 1$	$e^{\ln x} = x$

🔍 **Evaluate** expressions involving natural logarithms using the log laws.

a	$\ln e^{10}$	**b**	$e^{\ln 10}$	**c**	$\ln 5 + \ln 3$
d	$\ln \sqrt{e}$	**e**	$\ln e^3$	**f**	$\ln e^{1.3}$
g	$e^{\ln\left(\frac{1}{5}\right)}$	**h**	$\ln 12 - \ln 4$	**i**	$\ln \sqrt[3]{e}$

1 Use the ln button on your calculator to evaluate the following expressions.

a $\ln 1$	**b** $\ln 3$	**c** $\ln \frac{1}{8}$	**d** $\ln e^2$
0	1.099	-2.079	2

2 Evaluate these logarithms without a calculator.

a $\ln e^3$	**b** $\ln e^{-2}$	**c** $e^{\ln 2}$	**d** $e^{\ln 10}$
3	–2	2	10
e $\ln e^{\frac{1}{5}}$	**f** $e^{\ln \frac{1}{2}}$	**g** $e^{\ln(4^2)}$	**h** $e^{\ln \frac{1}{5}}$
$\frac{1}{5}$	$\frac{1}{2}$	16	$\frac{1}{5}$

3 Evaluate these expressions using the log law $\ln a^n = n \ln a$ and the fact that $\ln e = 1$

a $e \ln e$	**b** $\frac{1}{e} \ln \frac{1}{e}$	**c** $3 \ln e^2$
e	$-\frac{1}{e}$	6
d $\ln e^5$	**e** $\ln e^{200}$	**f** $\ln e^{-6}$
5	200	–6
g $\ln \frac{1}{e^6}$	**h** $\ln e^{-1}$	**i** $\ln e^{\frac{1}{2}}$
–6	–1	$\frac{1}{2}$
j $\ln \sqrt{e}$	**k** $\ln \frac{1}{\sqrt{e}}$	**l** $\ln \frac{1}{\sqrt{e^3}}$
$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{3}$
m $e \ln e^3 - e \ln e$	**n** $\ln e + \ln \frac{1}{e}$	**o** $e \ln e^3 - e \ln e$
$2e$	0	$2e$

3 An object is dropped from an aeroplane. Let s be the speed of the particle in metres per second, and t be time in seconds. The speed of the particle is given by the formula $s = 10(1 - e^{-t})$.

a Find the speed of the particle two seconds after release.

8.65 m/s

b How long does it take for the particle to reach a speed of 5 metres per second?

$t = \ln 2 = 0.69$ seconds

c Does this particle ever reach a speed of 10 metres per second? Justify your answer.

No, it approaches but never reaches 10.

4 When antibiotics are introduced into the human body, the concentration the antibiotic in the bloodstream can be modelled by the equation

$$C(t) = 9(e^{-0.2t} - e^{-0.4t})$$

where t is time in hours and C is measured in $\mu g\ mL^{-1}$.

a What is the concentration of the antibiotic after 1 hour?

1.34 μgmL^{-1}

b Let $x = e^{-0.2t}$ and show that when $C = 2$, the resultant quadratic is $9x^2 - 9x + 2 = 0$.

Let $x = e^{-0.2t}$, then $9(e^{-0.2t} - e^{-0.4t}) = 9(x - x^2)$

When $C = 2$:

$2 = 9(x - x^2)$

$2 = 9x - 9x^2$

$9x^2 - 9x + 2 = 0$

c Solve for x, and hence show that it will take just over 2 hours to reach $2\mu g\ mL^{-1}$.

$(3x - 1)(3x - 2) = 0$

$\therefore e^{-0.2t} = \frac{1}{3}, \frac{2}{3}$

$t_1 = -ln(2/3)/0.2 = 2.03$ hours

$t_2 = -ln(1/3)/0.2 = 5.49$ hours

d You should have acquired two solutions. What could be a biological reason as to why there may be two times when the concentration is $2\mu g\ mL^{-1}$?

The concentration rises initially as the antibiotic is absorbed, reaches a peak, and then falls as it is metabolised.

This concentration occurs twice: once while rising and the other while falling

24 Solve:

a $6x^2 + 7x - 5 = 0$

$x = \frac{1}{2}$ or $x = -\frac{5}{3}$

b $6\sin^2 x + 7\sin x - 5 = 0$ for $x \in [0, 2\pi]$

Let $u = \sin x$:

$u = \frac{1}{2}$ or $u = -\frac{5}{3}$

Since $-1 \leq \sin x \leq 1$, only $\sin x = \frac{1}{2}$ is valid

$x = \frac{\pi}{6}$ or $x = \frac{5\pi}{6}$

c $6(\log_3 x)^2 + 7\log_3 x - 5 = 0$

Let $y = \log_3 x$: $6y^2 + 7y - 5 = 0$

$y = \frac{1}{2}$ or $y = -\frac{5}{3}$

$\log_3 x = \frac{1}{2}$

$x = 3^{\frac{1}{2}} = \sqrt{3}$

$\log_3 x = -\frac{5}{3}$

$x = 3^{-\frac{5}{3}} = \frac{1}{\sqrt{3^5}} = \frac{1}{9\sqrt{3}}$

d $6e^{2x} + 7e^x - 5 = 0$

Let $z = e^x$: $6z^2 + 7z - 5 = 0$

$z = \frac{1}{2}$ or $z = -\frac{5}{3}$

Since $e^x > 0$, only $e^x = \frac{1}{2}$ is valid

$x = \ln\left(\frac{1}{2}\right) = -\ln 2$

Exponential Rate of Change Problems involving Logarithms

Example Solve exponential rate of change problems involving logarithms

The number B of bacteria in a laboratory culture is growing according to $B = 4000e^{0.1t}$, where t is the time in hours after the experiment was started.
Answer these questions correct to three significant figures:

a How many bacteria are there initially?

$B(0) = 4000e^0 = 4000$

b How many bacteria are there after 5 hours?

$B(5) = 4000e^{0.1(5)} \approx 6590$

c Find the rate at which the number of bacteria is changing.

$\frac{dB}{dt} = 400e^{0.1t}$

d At what rate are the bacteria increasing after 5 hours?

Let $t = 5$

$\frac{dB}{dt} = 400e^{0.5} = 518 \text{ bacteria/h}$

e When are there 10 000 bacteria?

$$B(t) = 10000$$
$$4000e^{0.1t} = 10000$$
$$e^{0.1t} = 2.5$$
$$0.1t = \ln 2.5$$
$$t = \frac{\ln 2.5}{0.1} = 9.16 \text{ hours}$$

f When are the bacteria increasing by 10 000 per hour?

$\frac{dB}{dt} = 10000$

$400e^{0.1t} = 10000$

$e^{0.1t} = 25$

$0.1t = \ln 25$

$t = \frac{\ln 25}{0.1} = 32.2 \text{ hours}$

Practice

The temperature of a metal in °C as it cools down over t minutes is $T = 27 + 219e^{-0.032t}$
Answer these questions correct to three significant figures:

a What is the temperature initially?

246° C

b What temperature will the rod cool to eventually?

27°C

c Find the rate at which the temperature is changing.

$-7.01e^{-0.032t}$ °C/min

d At what rate is the temperature changing after 20 minutes?

–3.68 °C/min

e When is the temperature 200°C?

7.37 min

f When is the metal cooling at 5°/min?

10.6 minutes

1 A population P of insects over time t weeks is given by $P = 3e^{1.4t} + 12\,569$.

a What is the initial population?

$At\ t = 0: P = 3e^0 + 12\,569 = 3 + 12\,569 = 12\,572$

b Find the rate of change, to the nearest whole number, of the number of insects after:

i 3 weeks

ii 7 weeks

$\frac{dP}{dt} = 3(1.4)e^{1.4t} = 4.2e^{1.4t}$

$i\ At\ t = 3: \frac{dP}{dt} = 4.2e^{4.2} \approx 280$ insects/week

$ii\ At\ t = 7: \frac{dP}{dt} = 4.2e^{9.8} \approx 75742$ insects/week

2 The displacement of a particle over time t seconds is given by $x = 2e^{4t}$ metres.

a What is the initial displacement?

At t = 0: $x = 2e^0$ = 2 m

b What is the exact velocity after 10 s?

$\frac{dx}{dt} = 8e^{4t}$

$at\ t = 10: v = 8e^{40}$ m/s

c Find the acceleration after 2 s correct to 1 decimal place.

$\frac{dv}{dt} = 32e^{4t}$

$at\ t = 2: a = 32e^8 \approx 95\,390.7$ m/s^2

3 The displacement of an object in cm over time t seconds is given by $x = 6e^{-0.34t} - 5$. Find:

a the initial displacement

$At\ t = 0: x = 6e^0 - 5 = 1$ cm

b the initial velocity

$\frac{dx}{dt} = 6(-0.34)e^{-0.34t} = -2.04e^{-0.34t}$

$at\ t = 0: v = -2.04$ cm/s

c the displacement after 4 s

–3.46 m

d the velocity after 9 s

–0.096 cm/s

e the acceleration after 2 s

$\frac{dv}{dt} = 0.694e^{-0.34t}$

At $x = 2, a = 0.35$ cm/s²

4 A population P is varying with time t in hours according to the formula $P = 2000\,e^{0.3t}$.

a Find the population when $t = 5$.

$P(5) = 8963$

b Find, as a function of t, the rate $\frac{dP}{dt}$ the population is changing, and show $\frac{dP}{dt} = 0.3P$.

$\frac{dP}{dt} = 600e^{0.3t} = 0.3 \times 2000e^{P} = 0.3P$

c Find the rate at which the population is changing when $t = 5$.

$P'(t) = 2689$ individuals/hr

d Find the average rate of change over the first 5 hours.

$\frac{P(5) - P(0)}{5 - 0} = 1393$ individuals/hr

5 The concentration $C = 2000\,e^{-2t}$ of a chemical is varying with time t in years.

a Find the concentration when $t = 2$.

36.63

b Find, as a function of t, the rate $\frac{dC}{dt}$ at which the concentration is changing.

$\frac{dC}{dt} = -4000e^{-2t}$

c Find the rate at which the concentration is changing when $t = 2$.

-73.26/yr

6 The price $\$P$ of an item is rising with inflation according to the formula $P = 150\,e^{0.04t}$, where t is the time in years.

a When will the price be \$300?

$t = 25 \ln 2 \approx 17.33$ yrs

b Find the rate $\frac{dP}{dt}$ at which the price is rising.

$\frac{dP}{dt} = 6e^{0.04t}$

c When will the price be rising at \$300 per year?

$t = 25 \ln 50 \approx 97.80$ yrs

7 The population P of cats on Snake Island is $P = 1000\,e^{0.4t}$, where t is the time in years after time zero when the population was first estimated.

a Find the rate at which the population is increasing.

$\frac{dP}{dt} = 400e^{0.4t}$

b Find the population, and the rate of increase, when $t = 5$, both correct to 2 s.f.

$P(5) \approx 7400 \text{ cats}, P'(5) \approx 3000 \text{ cats/yr}$

c When will the cat population reach 20 000 cats, correct to one decimal place?

$t = \frac{5}{2}\ln 20 \approx 7.5 \text{ yrs}$

d When will the rate of increase reach 20 000 cats per year, correct to one decimal place?

$t = \frac{5}{2}\ln 50 \approx 9.8 \text{ yrs}$

8 A radioactive substance decays in such a way that the mass M of a sample t years after it has been measured and stored is $M = M_0e^{-0.1t}$, where M_0 is its original mass.

a How long until half the sample is gone?

Let $M = \frac{M_0}{2}$

$\frac{1}{2} = e^{-0.1t}$

$\ln\frac{1}{2} = -0.1t$

$t = -10\ln\frac{1}{2} \approx 6.931 \text{ yrs}$

b Find the rate $\frac{dM}{dt}$ at which the sample is decaying, and show that $\frac{dM}{dt} = -0.1 \times M$.

$\frac{dM}{dt} = -0.1 \times M_0e^{-0.1t} = -0.1M$

c What percentage of the remaining sample decays each year?

Every year you multiply by $e^{-0.1}$

This is a multiplier of $\times 0.9048\ldots$

$\therefore$ 9.516% decrease

d When will the rate of decay be 1% of the original mass M_0 per year?

Let $\frac{dM}{dt} = -0.01M_0$

$t = -10\ln 10 \approx 23.03 \text{ yrs}$